Predicting Whether a Tumour Is Cancerous Using Machine Learning

Predictive Analytics Project

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Project type: Predictive Analytics

Data used: Wisconsin Breast Cancer dataset (569 patients, 30 measurements)

Tool used: Python

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2. Abstract

569	30	97.4%	92.9%
patients in the data	measurements each	correct predictions	cancers found

This project uses machine learning to predict whether a tumour is **malignant** (cancerous) or **benign** (not cancerous). The data has 569 patients. For each patient, 30 measurements of cell nuclei are available.

Three models were built and compared: Logistic Regression, Random Forest and Gradient Boosting. The data was split into two parts. 80% was used to teach the models and 20% (114 patients) was kept hidden to test them fairly.

All three models were above 96% accurate. After tuning, the best model (a Random Forest) was correct for 97.4% of the test patients. It found 39 of the 42 cancer cases and raised no false alarms. The most useful measurements were the **size** and **irregular shape** of the cell nuclei.

The same method is used by banks to predict whether a customer will repay a loan, so the project also explains how this works for finance.

> **In simple words:** We taught a computer to look at 30 numbers about a tumour and say if it is cancer or not. It got about 97 out of 100 right.

3. Introduction

What is predictive analytics? It means using past data to guess what will happen in the future or for a new case. For example, a bank looks at old loan records to guess if a new customer will repay.

What is machine learning? It is a way of letting a computer learn patterns from examples instead of writing fixed rules. We show the computer many cases where the answer is already known. It learns the pattern and then predicts the answer for new cases.

What is classification? When the answer is a category (Yes/No, Cancer/Not cancer, Default/No default), the task is called classification. This project is a classification problem with two categories.

Objectives of the project

- Build at least three machine learning models to classify tumours.
- Test them fairly on patients the model has never seen.
- Find which model is best and improve it.
- Find which measurements matter most.
- Understand how the same idea applies in finance.

> **In simple words:** The project is like training a student with solved examples, then giving a surprise test to see how well they learned.

4. Problem Statement

Doctors need to decide if a lump is cancerous. A wrong decision is costly. If a cancer is missed, treatment is delayed. If a healthy person is called cancerous, they face stress and extra tests.

Problem: Can we use the measurements of a cell sample to predict automatically and accurately whether a tumour is malignant or benign?

How the model works

INPUT		MODEL		OUTPUT
30 measurements (size, shape, texture)	->	learns from 455 known patients	->	Malignant or Benign

What counts as success?

- **High accuracy:** most predictions should be correct.
- **High recall:** almost every real cancer should be caught. This is the most important goal.
- **High precision:** when the model says cancer, it should usually be right.

In simple words: Missing a real cancer is worse than a false alarm, so catching all real cases matters most.

5. Data Discussion

Source: The Wisconsin Breast Cancer Diagnostic dataset (built into scikit-learn). **Size:** 569 patients and 30 measurements each. **Target:** the diagnosis: 212 malignant (37.3%) and 357 benign (62.7%). There are 0 missing values and 0 duplicate rows.

Ten basic measurements were taken from an image of the cells. For each, three values were recorded: the **mean**, the **standard error** and the **worst** (largest) value. So 10 x 3 = 30 features.

Measurement	What it means in plain words
radius	Distance from the centre of the cell nucleus to its edge
texture	How uneven the grey shades are
perimeter	Length of the boundary of the nucleus
area	Size of the nucleus
smoothness	How smooth the edge is
compactness	How tightly packed the shape is
concavity	How deep the dents in the edge are
concave points	Number of dents in the edge
symmetry	How balanced the shape is
fractal dimension	How rough or wavy the boundary is

Excel data (sample of 8 rows)

ID	Diagnosis	mean radius	mean texture	mean perimeter	mean area	mean smoothness	mean concavity
1	Malignant	18.0	10.4	122.8	1001.0	0.118	0.300
2	Malignant	20.6	17.8	132.9	1326.0	0.085	0.087
3	Malignant	19.7	21.2	130.0	1203.0	0.110	0.197
4	Malignant	11.4	20.4	77.6	386.1	0.142	0.241
20	Benign	13.5	14.4	87.5	566.3	0.098	0.067
21	Benign	13.1	15.7	85.6	520.0	0.107	0.046
22	Benign	9.5	12.4	60.3	273.9	0.102	0.030
23	Malignant	15.3	14.3	102.5	704.4	0.107	0.208

Excel file: Predictive_BreastCancer_Data.xlsx | Sheet: Data (569 rows, 33 columns in total)

In simple words: The data is clean, so we can use it directly. The full data is in the Excel file Predictive_BreastCancer_Data.xlsx.

6. Exploratory Analysis

Before building models we look at the data to understand it. This step is called exploratory analysis.

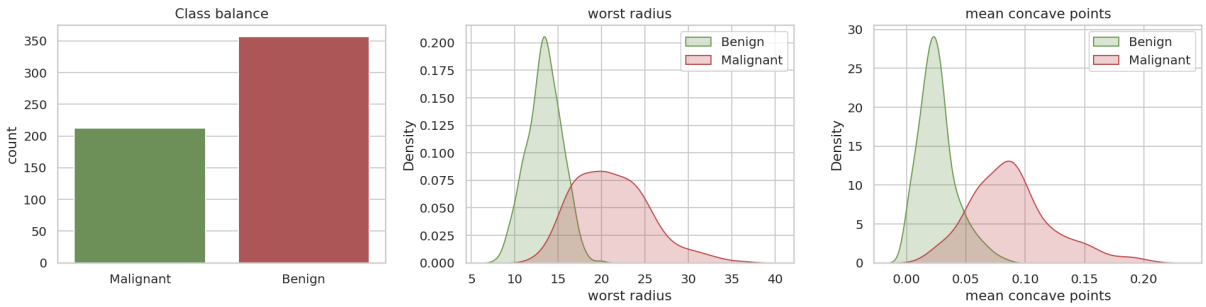


Figure 1: Left: number of benign and malignant cases. Middle and right: how two measurements differ between the two groups

The left chart shows more benign cases than malignant ones. The other two charts show that malignant tumours (red) usually have larger values than benign ones (green) with only a small overlap. That is a good sign: it means the measurements can help separate the two groups.

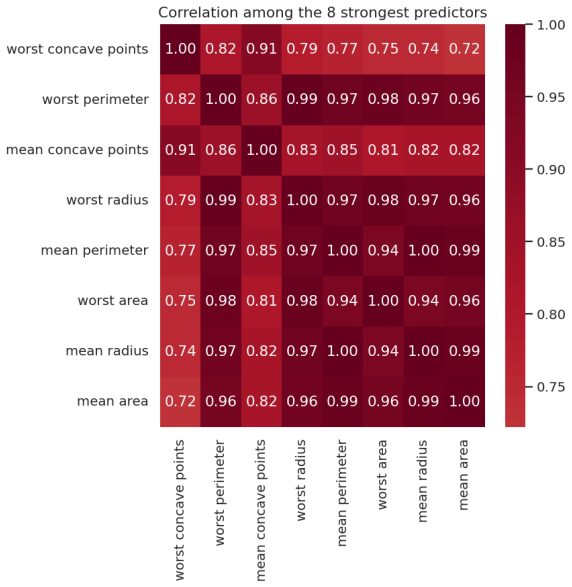


Figure 2: Relationship between the 8 measurements most linked to cancer (dark red = move together)

Strongest links with cancer: **worst concave points** (0.79), **worst perimeter** (0.78), **mean concave points** (0.78). Size measurements move together, which is expected: a bigger radius also means a bigger area and perimeter.

7. Method, Tools and Technologies

Tools used

Tool	Used for
Python	Loading and checking the data, charts, machine learning models, and creating the Excel file and this report

Methods (models) used

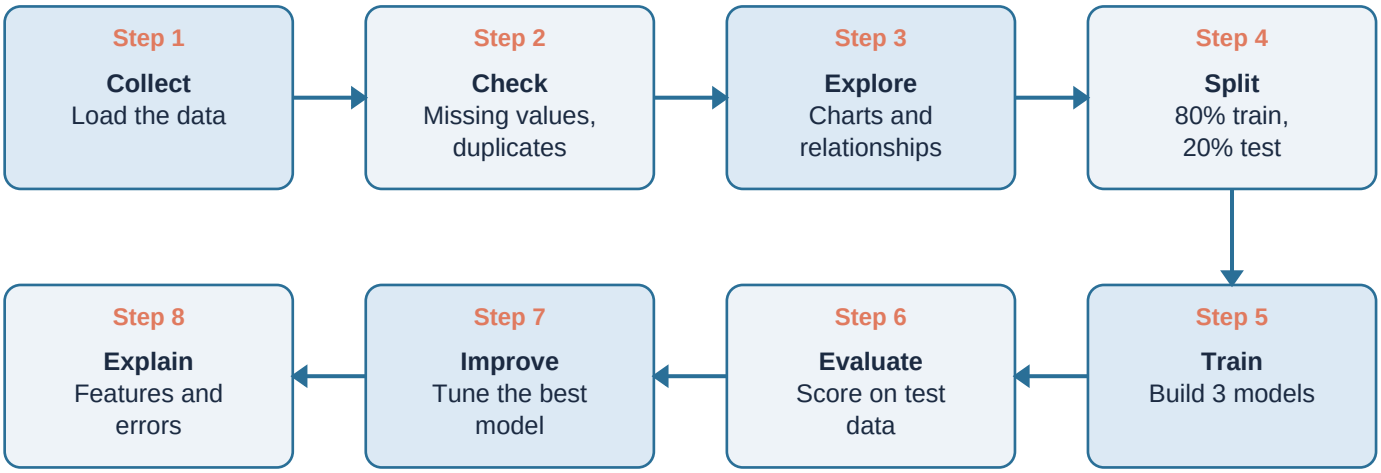
Model	How it works in simple words
Logistic Regression	Draws a smart dividing line between the two groups. Simple and easy to explain.
Random Forest	Builds many decision trees (like flow charts of yes/no questions) and takes a vote.
Gradient Boosting	Builds trees one after another, each one fixing the mistakes of the previous one.

Methods for checking the models

- **Train-test split:** keep 20% of the data hidden for a fair final test.
- **Cross-validation:** test the model 5 times on different parts of the data to be sure the result is not luck.
- **Grid search:** try many settings automatically and pick the best.
- **Accuracy, precision, recall, F1, ROC-AUC:** scores explained in the Results pages.

8. Workflow

The project followed these steps in order. Each step uses the output of the previous one.



What happens at each step

- **Collect and check:** make sure the data is complete and correct.
- **Explore:** understand the patterns before modelling.
- **Split:** hide some data so the final test is fair.
- **Train and evaluate:** the models learn from the 80% and are scored on the hidden 20%.
- **Improve:** adjust model settings to catch more cancers.
- **Explain:** find which measurements matter and where mistakes happen.

> **In simple words:** It is like preparing for an exam: study (train), take a mock test (evaluate), fix weak areas (tune), then review mistakes (error analysis).

9. Procedure

1. Loaded the dataset from scikit-learn and changed the label so that 1 means malignant (the case we want to catch).
2. Checked for missing values and duplicates. Found none, so no cleaning was needed.
3. Split the data: 455 patients for training and 114 for testing. The split keeps the same share of cancers in both parts (stratified).
4. Standardised the measurements for Logistic Regression so all of them are on the same scale.
5. Trained Logistic Regression, Random Forest and Gradient Boosting on the training data.
6. Tested each model on the hidden test data and also with 5-fold cross-validation.
7. Tuned the Random Forest with grid search (best: 400 trees, depth 5, minimum leaf size 3).
8. Drew confusion matrices, ROC curves and feature importance, and tested different decision thresholds.

Core code

```
X, y = load_breast_cancer(as_frame=True, return_X_y=True); y = 1 - y
Xtr, Xte, ytr, yte = train_test_split(X, y, test_size=.2, stratify=y, random_state=42)
rf = RandomForestClassifier(n_estimators=300, random_state=42).fit(Xtr, ytr)
print(accuracy_score(yte, rf.predict(Xte)))
grid = GridSearchCV(RandomForestClassifier(random_state=42),
        {'n_estimators':[200,400], 'max_depth':[None,5,10], 'min_samples_leaf':[1,3]},
        cv=5, scoring='recall').fit(Xtr, ytr)
```

The full code with all outputs is in the notebook **Predictive_BreastCancer_Classification.ipynb**.

10. Results: Model Comparison

All models were scored on the same 114 hidden test patients.

Model	Accuracy	Precision	Recall	F1	ROC-AUC	CV Acc.
Logistic Regression	0.965	0.975	0.929	0.951	0.996	0.974
Random Forest	0.974	1.000	0.929	0.963	0.994	0.953
Gradient Boosting	0.965	1.000	0.905	0.950	0.995	0.951
Tuned Random Forest	0.974	1.000	0.929	0.963	0.995	-

Excel file: Predictive_BreastCancer_Data.xlsx | Sheet: Model_Results

What the scores mean

- **Accuracy:** out of 100 patients, how many were predicted correctly.
- **Precision:** when the model says cancer, how often it is right.
- **Recall:** out of all real cancers, how many the model found.
- **F1:** one score that balances precision and recall.
- **ROC-AUC:** how well the model ranks cancers above non-cancers (1.0 is perfect).
- **CV Acc.:** average accuracy over 5 different test rounds.

Which model is best?

All models scored above 96% accuracy and above 0.99 ROC-AUC. Random Forest had the highest test accuracy (0.974). In cross-validation, Logistic Regression was the most steady (0.974). The tuned Random Forest was chosen as the final model because it had no false alarms and the best recall among the tree models (0.929).

In simple words: The simple model is almost as good as the complex ones, so the data is easy to separate. The gaps between models are tiny.

11. Confusion Matrices and ROC Curves

A **confusion matrix** counts correct and wrong predictions. A **ROC curve** shows how well the model separates the two groups.

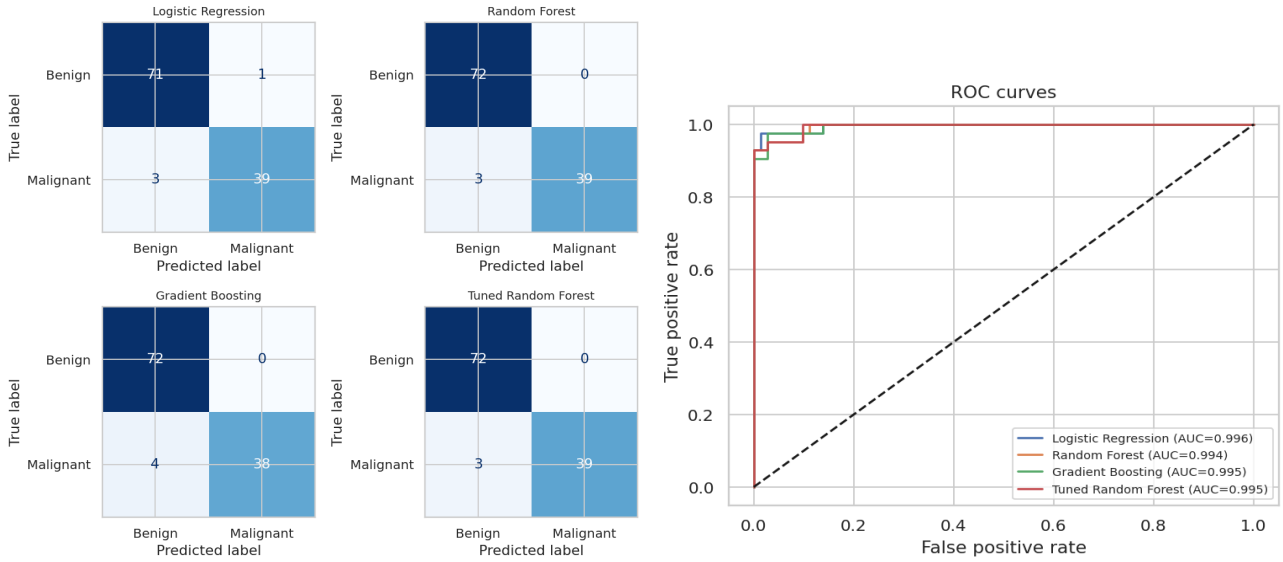


Figure 3 (left): confusion matrices of the four models. Figure 4 (right): ROC curves; closer to the top-left corner is better.

	Meaning
True negative	Healthy, predicted healthy (correct)
False positive	Healthy, predicted cancer (false alarm)
False negative	Cancer, predicted healthy (missed case)
True positive	Cancer, predicted cancer (correct)

The tuned model made 0 false alarms and missed 3 cancers out of 42. All curves are very close to the top-left corner (AUC about 0.99), which means all models separate the two groups very well.

12. Feature Importance

Feature importance tells us which measurements helped the model most in making its decision.

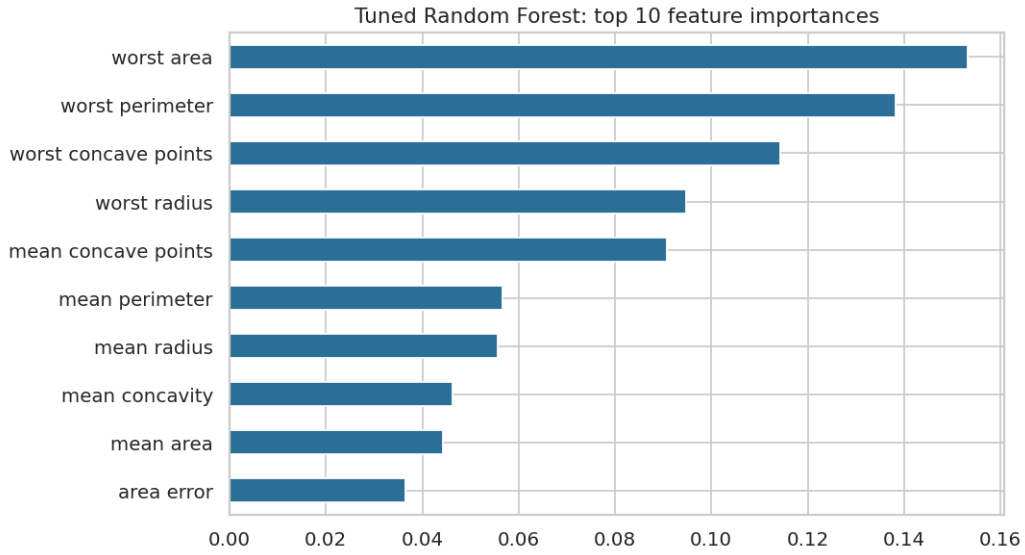


Figure 5: Ten most important measurements in the tuned Random Forest

Top five measurements

Rank	Measurement	Importance
1	worst area	0.153
2	worst perimeter	0.138
3	worst concave points	0.114
4	worst radius	0.095
5	mean concave points	0.091

Excel file: Predictive_BreastCancer_Data.xlsx | Sheet: Feature_Importance

The top features are about **size** (worst area, perimeter and radius) and **irregular shape** (concave points). In simple terms, bigger cells with more uneven edges are more likely to be cancerous.

> **In simple words:** The model mainly asks: how big is the cell and how rough is its edge? This agrees with medical thinking.

13. Error Analysis

72	0	3	39
True negatives	False positives	False negatives	True positives

On the 114 hidden patients, the tuned model was right 111 times. It never gave a false alarm. It missed 3 real cancers. In medicine and in banking, a missed case (false negative) is usually the most expensive mistake.

Can we catch more cancers?

The model gives a probability of cancer. Normally we say cancer if the probability is 50% or more. If we lower this limit (threshold), the model flags more people. The table shows what happens.

Threshold	Accuracy	Precision	Recall	False alarms	Missed cancers
0.5	0.974	1.000	0.929	0	3
0.4	0.965	0.975	0.929	1	3
0.3	0.965	0.975	0.929	1	3
0.2	0.921	0.851	0.952	7	2
0.1	0.886	0.764	1.000	13	0

Excel file: Predictive_BreastCancer_Data.xlsx | Sheet: Threshold_Analysis

At threshold 0.1, no cancer is missed, but there are 13 false alarms and accuracy drops to 88.6%. The right choice depends on how costly each mistake is.

In simple words: Lowering the limit is like a smoke alarm that is more sensitive: it never misses a fire but rings more often for nothing.

14. Insights (Observations)

1. The data was clean and had 569 patients, so no cleaning was needed.
2. All three models were above 96% accurate, so these measurements contain a strong signal about cancer.
3. The tuned Random Forest gave 97.4% accuracy, 100% precision and 92.9% recall on unseen patients.
4. Simple Logistic Regression was almost as good as the complex models. A simple model can be enough.
5. Size and shape measurements (worst area, worst perimeter, concave points) are the strongest predictors.
6. The model missed 3 cancers and had no false alarms. Lowering the threshold can remove misses but adds false alarms.
7. The differences between models are very small on only 114 test patients. One extra mistake changes accuracy by about 0.9 percentage points.

In simple words: The most important lesson is that accuracy alone is not enough. We must also look at which kind of mistake the model makes.

15. Relevance to Finance

This project is about medicine, but the method is exactly the one used in banking and finance. Only the data changes.

In this project	In finance (loan default)
Patient	Loan customer
Cell measurements	Income, loan amount, credit score, past repayments
Malignant (1)	Default / bad loan (1)
Benign (0)	Repaid / good loan (0)
Missed cancer (false negative)	Approving a loan that later defaults
False alarm (false positive)	Rejecting a good customer
Recall	Share of bad loans caught
Threshold	Score cut-off for approving loans

Where else is this used?

- Credit scoring and loan approval
- Fraud detection in cards and online payments
- Predicting customer churn
- Insurance claim risk
- Bankruptcy and stock-trend prediction

In simple words: If we replace the tumour data with loan data, the same code can predict who may default.

16. Limitations

- **Small data:** only 569 patients from one source. The result may not hold for other hospitals or groups.
- **Small test set:** only 114 test patients, so the scores are approximate.
- **Not a doctor:** the model supports decisions but cannot replace medical experts.
- **Association, not cause:** feature importance shows what is linked with cancer, not what causes it.
- **No outside check:** the model was not tested on a completely new dataset.
- **Not a finance dataset:** the finance relevance is explained by analogy, not tested on loan data.

> **In simple words:** The results are encouraging but they come from a small, single dataset. They should be checked on more data before any real use.

17. Recommendations / Suggestions

1. **Use the model as a helper, not a judge.** Doctors (or loan officers) should make the final decision.
2. **Choose the threshold by cost.** If missing a case is very costly, use a lower threshold.
3. **Collect more data** from different places and test the model again.
4. **Try more models** such as SVM or XGBoost and compare them.
5. **Check fairness.** Make sure the model works equally well for different groups.
6. **For finance:** repeat this project on a real loan dataset (for example, the Kaggle Loan Prediction data).
7. **Monitor the model** regularly. If the data changes over time, retrain it.

18. Conclusion

This project built and compared three machine learning models to predict whether a tumour is malignant or benign. All models were more than 96% accurate. After tuning, the Random Forest correctly classified 97.4% of unseen patients, with no false alarms, and found 39 of 42 cancers.

The most important measurements are the size and shape of the cell nuclei. The project also showed that the type of mistake matters as much as accuracy, and that the decision threshold can be changed to catch more real cases at the cost of more false alarms.

The same steps (clean, split, train, test, tune, explain) can be used in finance to predict loan default, fraud or customer churn. This makes the project a useful practice for a future finance and analytics career.

What I learned

- How to prepare data and split it fairly for testing.
- How to train, compare and tune classification models.
- How to read confusion matrices, ROC curves and feature importance.
- How to link a technical result to a business decision.

19. References

1. Wolberg, W. H., Street, W. N. and Mangasarian, O. L. (1995). Breast Cancer Wisconsin (Diagnostic) Data Set. UCI Machine Learning Repository.
2. Pedregosa, F. et al. (2011). Scikit-learn: Machine Learning in Python. Journal of Machine Learning Research, 12.
3. Breiman, L. (2001). Random Forests. Machine Learning, 45(1), 5-32.
4. Friedman, J. H. (2001). Greedy Function Approximation: A Gradient Boosting Machine. Annals of Statistics, 29(5).
5. Hastie, T., Tibshirani, R. and Friedman, J. (2009). The Elements of Statistical Learning. Springer.
6. McKinney, W. (2010). Data Structures for Statistical Computing in Python. Proceedings of the 9th Python in Science Conference.
7. Hunter, J. D. (2007). Matplotlib: A 2D Graphics Environment. Computing in Science and Engineering, 9(3).

In simple words: These are the sources for the dataset, the software libraries and the algorithms used in this project.

20. Appendix

A. Excel data (sample of 10 rows)

ID	Diagnosis	radius	texture	perimeter	area	smoothness	concavity
1	Malignant	18.0	10.4	122.8	1001.0	0.118	0.300
2	Malignant	20.6	17.8	132.9	1326.0	0.085	0.087
3	Malignant	19.7	21.2	130.0	1203.0	0.110	0.197
4	Malignant	11.4	20.4	77.6	386.1	0.142	0.241
5	Malignant	20.3	14.3	135.1	1297.0	0.100	0.198
20	Benign	13.5	14.4	87.5	566.3	0.098	0.067
21	Benign	13.1	15.7	85.6	520.0	0.107	0.046
22	Benign	9.5	12.4	60.3	273.9	0.102	0.030
23	Malignant	15.3	14.3	102.5	704.4	0.107	0.208
24	Malignant	21.2	23.0	137.2	1404.0	0.094	0.110

Excel file: Predictive_BreastCancer_Data.xlsx | Sheet: Data (569 rows, 33 columns in total; columns shown: mean values only)

B. Excel summary by diagnosis

Measurement	Benign mean	Malignant mean	Ratio
mean radius	12.147	17.463	1.44
mean texture	17.915	21.605	1.21
mean perimeter	78.075	115.365	1.48
mean area	462.790	978.376	2.11
mean concavity	0.046	0.161	3.49
mean concave points	0.026	0.088	3.42

Excel file: Predictive_BreastCancer_Data.xlsx | Sheet: Summary_By_Diagnosis (30 measurements in the file)

C. Other sheets in the Excel file

Model_Results, **Confusion_Matrix**, **Threshold_Analysis**, **Feature_Importance** and **Data_Dictionary**. The first four tables appear in Results pages 10, 12 and 13.

D. Tool used

The whole project was done in **Python**: data loading, charts, machine learning models, the Excel file (Predictive_BreastCancer_Data.xlsx) and this PDF report. Run the notebook in Google Colab; seed 42 gives the same numbers.